

Research Article

Paper Title

First Author^{1*}

¹Department Name, University Name, City, Country

Hypergraph RAG + sparse autoencoder, using quantum linear algebra (Complex / Hilbert space): to visualise concept drift

Abstract

Place abstract text here

Keywords: Place keywords here

1. INTRODUCTION

Place text here

2. LITERATURE REVIEW

[1] For semantic space modelling.

2.1. Phase 0: Problem Definition

2.1.1. *Define the drift problem in one paragraph*

2.1.2. *Define the drift problem in one paragraph*

This project visualises concept drift in a HyperRAG system operating over simple medical datasets. As the system answers a query using a HyperRAG network, a sparse autoencoder identifies the concepts it relies on, and complex (quantum-inspired) linear algebra gives each concept a magnitude and an angle in Hilbert space; magnitude for how strongly the concept is present, angle for its relation to other concepts. Plotting these across queries turns the similarities and differences between concepts into visible geometry: as the query changes, the relative angles between concepts open and close, and that movement is the drift. The aim is to make the system's shifting internal concepts legible in a way a real-valued, magnitude-only view cannot. We want to see how the system judges relations in a Hilbert space, and how those relations change over different queries.

2.1.3. *Identify target audience*

Researchers, engineers and artists.

2.1.4. *Define expected output (visualiser + baseline comparison)*

A 3D render (a projection of the high-dimensional Hilbert space) of concepts moving as queries change, alongside a baseline of the same concepts in real vector space for comparison.

2.1.5. Write success criteria

The Hilbert-space view (angle + magnitude) reveals relational structure or drift between concepts that the real-valued, magnitude-only baseline does not. Equivalently; degrading the Hilbert-space representation down to a real vector space visibly loses information, and the visualiser makes that lost information apparent.

2.1.6. Record assumptions / risks

Information loss and spurious concepts in both the real and Hilbert-space representations, and the risk of overfitting to a small dataset.

2.2. Phase 1: Hypergraph RAG

2.2.1. Hypergraph Store

This section takes our initial data and stores it as a hypergraph, each entity, is a node and n-ary of facts are hyperedges. This data structure is built using hypernetx. This section has no quantum linear algebra as this is the store of our preliminary data structure.

2.2.2. Embedding Layer

Where entities and relations become embeddings, embed these in a Hilbert space.

3. RESULTS AND DISCUSSION

Place text here

4. CONCLUSION AND FUTURE WORK

Place text here

5. DECLARATION

Conflict of Interest: Place text here

Author Contribution: Place text here

6. REFERENCES

- [1] Timo Aukusti Laine. “The Quantum LLM: Modeling Semantic Spaces with Quantum Principles”. In: *OA Journal of Applied Science and Technology* 3.2 (2025), pp. 1–13. doi: 10.48550/arXiv.2504.13202. arXiv: 2504.13202 [cs.AI]. URL: <https://doi.org/10.48550/arXiv.2504.13202>.